Hence by an application of the l'Hôpital Rule we get,

$$\lim_{x\to a}\left(\frac{x^n-a^n}{x-a}\right) = \lim_{x\to a}\left(\frac{n\times x^{n-1}}{1}\right)$$

$$= n\times a^{n-1}.$$

**Example 7.35**

Evaluate the limit $\lim_{x\to 0}\left(\frac{\sin mx}{x}\right)$.

**Solution**

If we directly substitute $x=0$ we get an indeterminate form $\frac{0}{0}$ and hence we apply the l'Hôpital's rule to evaluate the limit as,

$$\lim_{x\to 0}\left(\frac{\sin mx}{x}\right) = \lim_{x\to 0}\left(\frac{m\times\cos mx}{1}\right)$$

$$= m$$

The next example tells that the limit does not exist.

**Example 7.36**

Evaluate the limit $\lim_{x\to 0}\left(\frac{\sin x}{x^2}\right)$.

**Solution**

If we directly substitute $x=0$ we get an indeterminate form $\frac{0}{0}$ and hence we apply the l'Hôpital's rule to evaluate the limit as,

$$\lim_{x\to 0^+}\left(\frac{\sin x}{x^2}\right) = \lim_{x\to 0^+}\left(\frac{\cos x}{2x}\right) = \infty$$

$$\lim_{x\to 0^-}\left(\frac{\sin x}{x^2}\right) = \lim_{x\to 0^-}\left(\frac{\cos x}{2x}\right) = -\infty$$

As the left limit and the right limit are not the same we conclude that the limit does not exist.

**Remark**

One may be tempted to use the l'Hôpital's rule once again in $\lim_{x\to 0^+}\left(\frac{\cos x}{2x}\right)$ to conclude

$$\lim_{x\to 0^+}\left(\frac{\cos x}{2x}\right) = \lim_{x\to 0^+}\left(\frac{-\sin x}{2}\right) = 0.$$

which is not true because it was not an indeterminate form.

**Example 7.37**

If $\lim_{\theta\to 0}\left(\frac{1-\cos m\theta}{1-\cos n\theta}\right)=1$, then prove that $m=\pm n$.

**Solution**

As this is an indeterminate form $\left(\frac{0}{0}\right)$, using the l'Hôpital's Rule

$$\lim_{\theta\to 0}\left(\frac{1-\cos m\theta}{1-\cos n\theta}\right) = \lim_{\theta\to 0}\left(\frac{m\sin m\theta}{n\sin n\theta}\right)$$



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